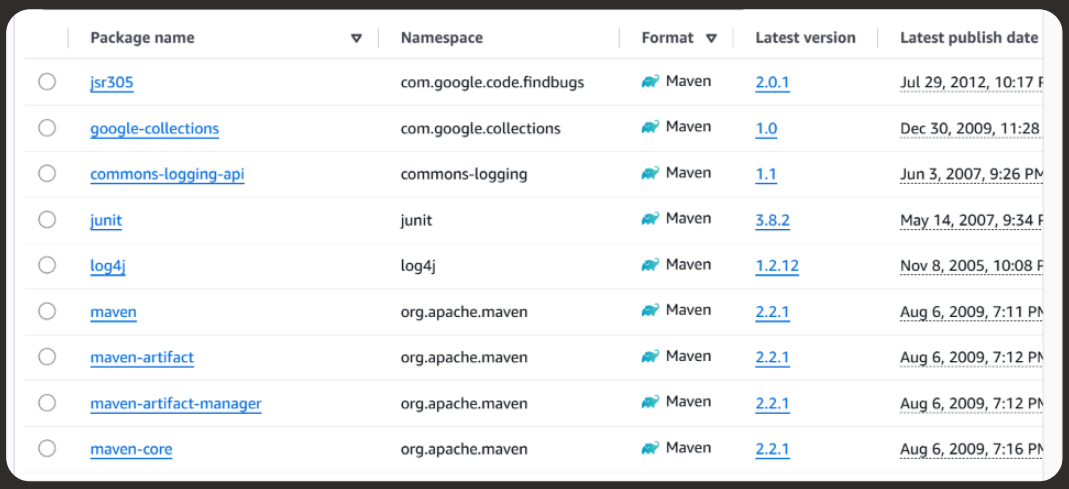
SECURE PACKAGES WITH CODEARTIFACT

By Sundram Singhal | September 2026



Package name	Namespace	Format	Latest version	Latest publish date
jsr305	com.google.code.findbugs	Maven	2.0.1	Jul 29, 2012, 10:17 F
google-collections	com.google.collections	Maven	1.0	Dec 30, 2009, 11:28
commons-logging-api	commons-logging	Maven	1.1	Jun 3, 2007, 9:26 PM
junit	junit	Maven	3.8.2	May 14, 2007, 9:34 F
log4j	log4j	Maven	1.2.12	Nov 8, 2005, 10:08 F
maven	org.apache.maven	Maven	2.2.1	Aug 6, 2009, 7:11 PM
maven-artifact	org.apache.maven	Maven	2.2.1	Aug 6, 2009, 7:12 PM
maven-artifact-manager	org.apache.maven	Maven	2.2.1	Aug 6, 2009, 7:12 PM
maven-core	org.apache.maven	Maven	2.2.1	Aug 6, 2009, 7:16 PM

Introducing Today's Project!

In this project, I will build CI/CD pipelines from scratch that will automate my building and deployment. I'm doing this project to learn how AWS CodeArtifact works.

Key tools and concepts

Services I used were CodeArtifact, maven.Key concepts I learnt include is creating repository, set up the IAM role and then download all the required dependenices into our CodeArtifact.

Project reflection

This project took me approximately 2 hours... The most challenging part was download the dependencies at the later part. It was most rewarding to finally complete the session.

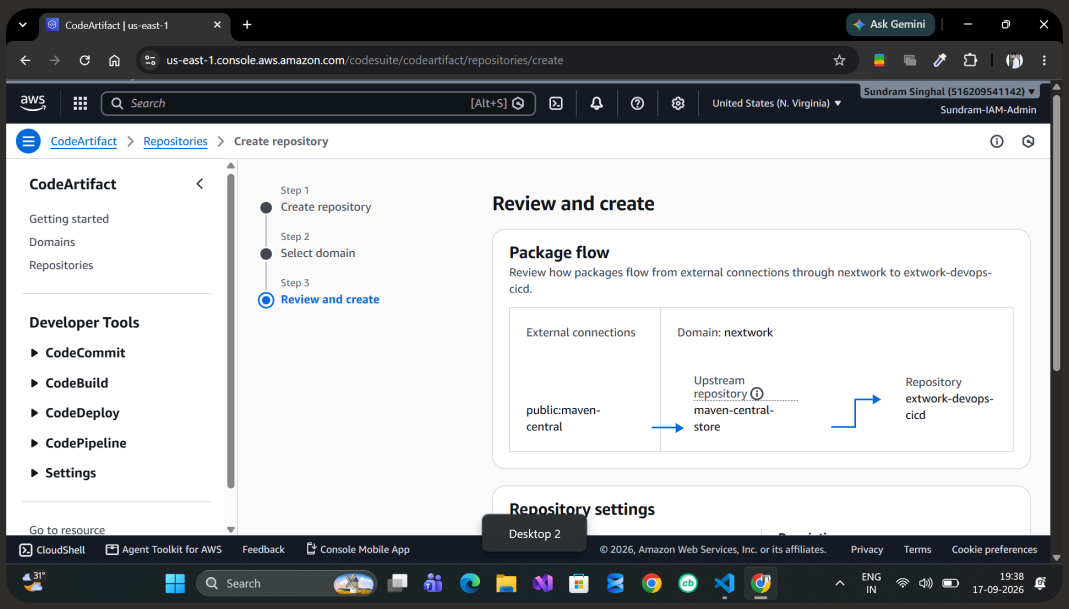
This project is part three of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project tomorrow itself.

CodeArtifact Repository

CodeArtifact is secure, central place to store all your software packages. Engineering teams use artifact repositories because it gives us a consistent, reliable place to store and retrieve these components. his gives you three big benefits: Security, Reliability and Control.

A domain is like a folder that holds multiple repositories belonging to the same project or organization. My domain is CodeArtifact domain.

A CodeArtifact repository can have an upstream repository, which means our backup repository which can be useful when we require packages that are not available in the primary repository. My repository's upstream repository is maven central which contains all the important libraries and also all the java developers push their libraries here.



CodeArtifact Security

Issue

To access CodeArtifact, we need to run the command in Step 3 to give your terminal a temporary password.I ran into an error when retrieving a token because it says "I don't know who you are, so I can't let you access CodeArtifact."

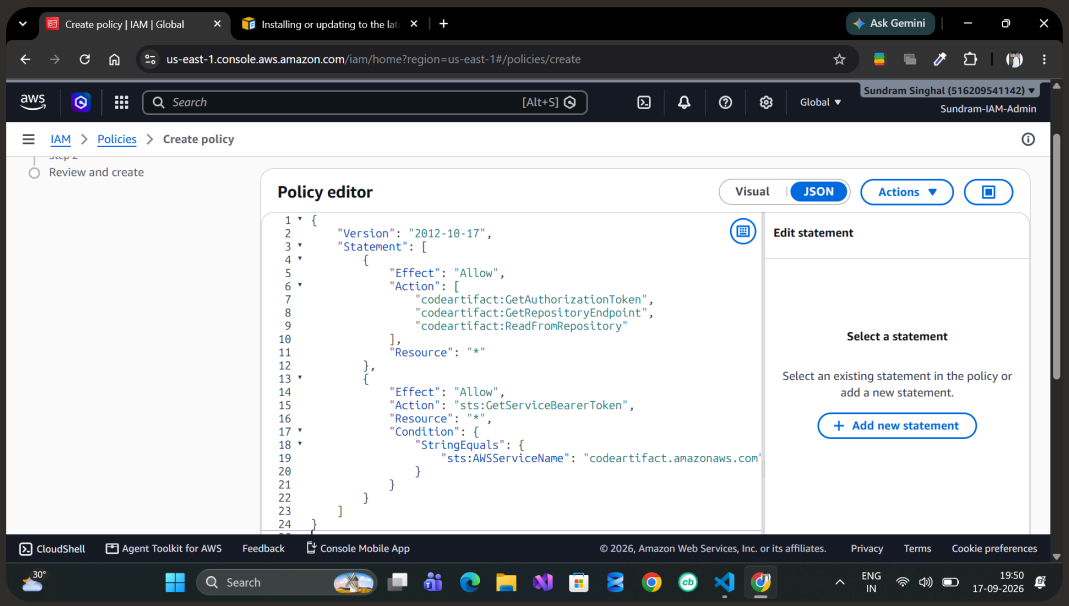
Resolution

To resolve the error with my security token, beasically i was restarting after 24 hours so i created a new token and then paste in vs code and i did not get any error.

It's security best practice to use IAM roles because IAM roles are a security best practice because they provide temporary credentials and allow applications to access AWS resources without storing long-term credentials, while supporting least-privilege access.

The JSON policy attached to my role

The JSON policy I set up grants (codeartifact:* actions) gives permission to get authentication tokens, find repository locations, and read packages from repositories.
The second part (sts:GetServiceBearerToken) allows temporarily elevated access specifically for CodeArtifact operations.

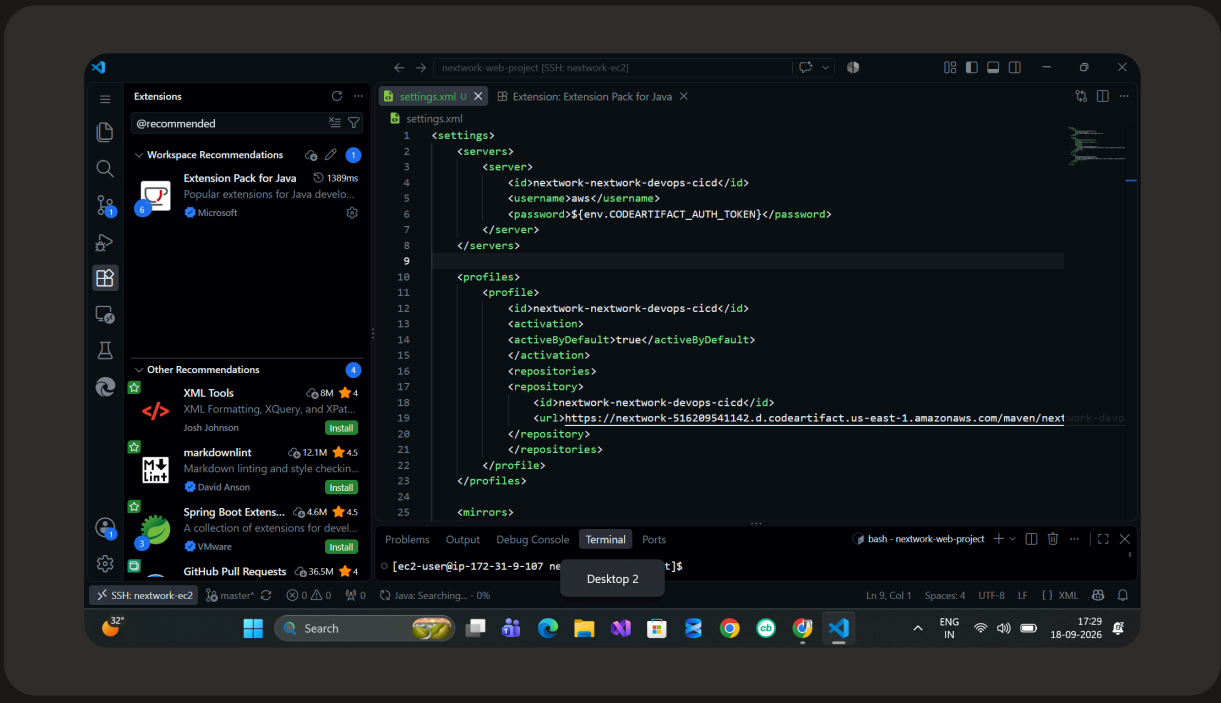


Maven and CodeArtifact

To test the connection between Maven and CodeArtifact, I compiled my web app using settings.xml

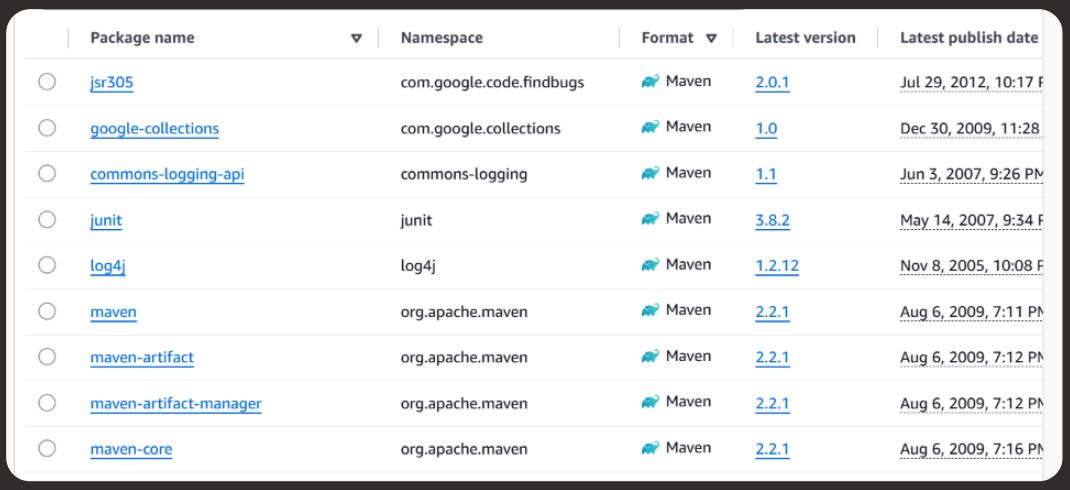
The settings.xml file configures Maven to work with codeartifact because we gave it the required information to authentication maven like token and rest of all the required stuffs.

Compiling means translating your project's code into a language that computers can understand and run. When you compile your project, you're making sure everything is correctly set up and ready to turn into a working app.



Verify Connection

After compiling, I checked nextwork-devops-cicd Repository. I noticed a list of so many packages there.



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Uploading My Own Packages

In a project extension, I also decided to complete secret mission which is to create my Secret Package. This is useful in situations where i want to share my code internally but wanna keep that private from the outside world.

To create my own package, I... I also generated a security hash because...To create my own package, I open the CloudShell terminal and then ran the required commands. I also generated a security hash because it helps me to protect my package from others, As it provides security.

To publish the package, When I look at the package details in CodeArtifact, I can see all the details about my newly added package there.

To validate my packages, I ran a command and then check the output to verify them.

